THE LAWS OF SOFTWARE TEAMS

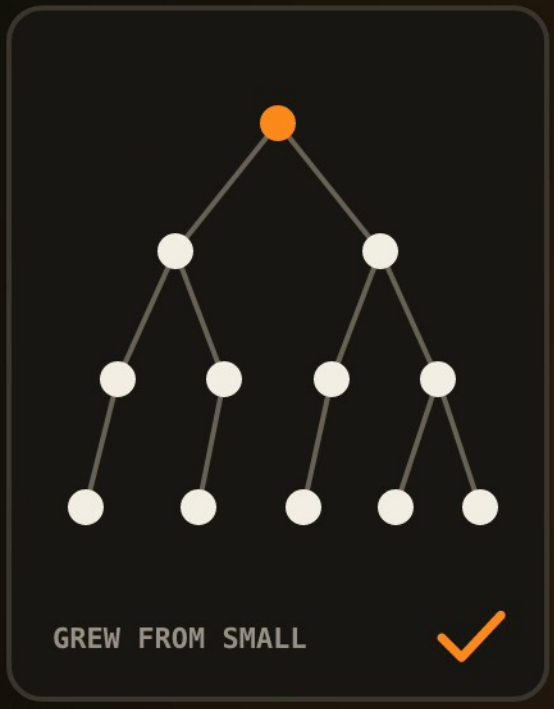
# Gall's law

## A complex system that works grew from a simple one that worked

swipe →

Ivan Rykov

# What works started small



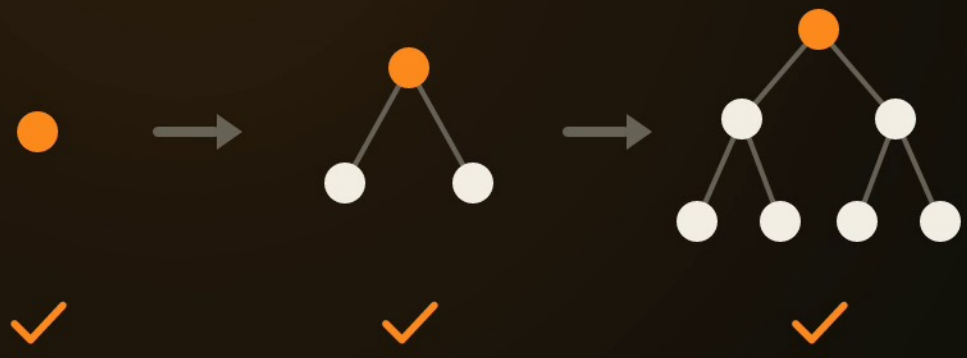

## Written down in 1975 by a children's doctor

illustration

# "A complex system that works is invariably found to have evolved from a simple system that worked"

John Gall · General Systemantics · 1975

a paediatrician who wrote about systems



each stage works before the next one · illustration

# Designed whole, never works

> "A complex system designed from scratch never works and cannot be patched up to make it work. You have to start over with a working simple system"
>
> John Gall, Systemantics, 1975, p. 71



the big design + patches



start over, small

# Linux:
# just a hobby

COMP.OS.MINIX · 25 AUGUST 1991

From: Linus Torvalds

“just a hobby, won’t be big and professional like gnu”



**10,000**

1991 · version 0.01

**23.3M+**

lines

2018 · version 4.15

circle area shows lines of code · 2018 count without comments

# Netscape: the full rewrite

rewrite from scratch

| 4.0 | ~~5.0~~ | 6.0 |
| --- | --- | --- |
| 1997 | never shipped | beta · April 2000 |

almost three years

JOEL SPOLSKY · APRIL 2000

> “the single worst strategic mistake that any software company can make”

Things You Should Never Do, Part I

# Nobody knows every requirement

KNOWN ON DAY ONE

- sign in
- create an invoice
- send it

A working system shows you what it really needs

FOUND BY USING IT

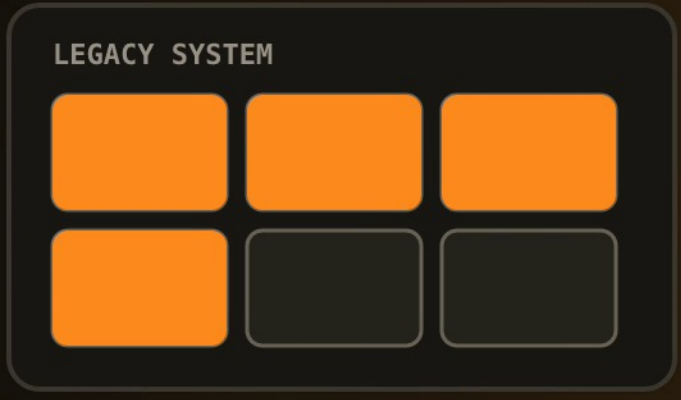
- time zones
- retries
- bulk edit
- audit log
- rate limits
- partial refunds
- export to CSV

illustration

# Start small
# Replace in pieces

UI | API | DB | deploy

the smallest version that works end to end — then grow it

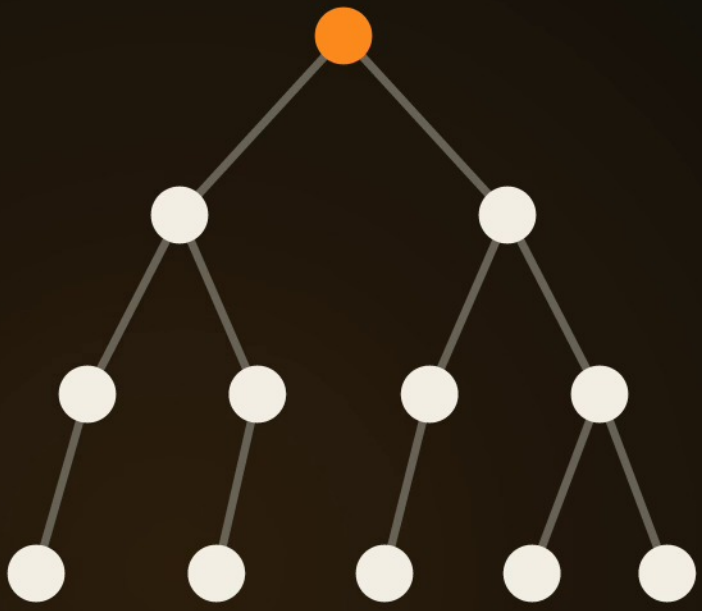



new pieces
around the old,
one at a time

the strangler fig pattern · Martin Fowler

08 TAKEAWAY

# Grow systems
# Do not design them whole



TAKEAWAY

# Grow systems
# Do not design them whole

Sources: John Gall, Systemantics, 1975 · Joel Spolsky, 2000 · Martin Fowler

next: Hyrum's law